Adaptive Traffic Signal Control Using Multi-Source Traffic State Prediction Under Urban Intelligent Transportation Systems

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Abstract

Accurate short-term traffic prediction is central to achieving effective adaptive traffic signal control (ATSC) in modern urban networks. To enhance predictive reliability, a multi-source fusion framework is introduced that integrates heterogeneous data from loop detectors, GPS probes, and video-based measurements into a unified spatio-temporal representation. A hybrid GCN–GRU model leverages this representation to forecast near-future traffic states, which in turn guide a model predictive control module that optimizes signal timings in real time. Experiments using a calibrated 15-intersection network show substantial performance gains, including a 23% reduction in average delay, a 28% decrease in maximum queue length, and a 29% improvement in throughput over fixed-time and rule-based adaptive systems. The results demonstrate that combining multi-source traffic prediction with predictive control yields more proactive and resilient operational strategies, offering a practical pathway for data-rich intelligent transportation systems.

Keywords

Adaptive traffic signal control, Multi-source traffic prediction, Intelligent Transportation Systems

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1 INTRODUCTION

Urban traffic congestion presents a significant challenge in modern cities, impacting economic productivity, environmental quality, and quality of life. Traditional fixed-time traffic signal control systems, operating on predetermined schedules regardless of real-time traffic fluctuations, frequently fail to manage the dynamic and complex nature of urban traffic flow effectively. These systems lack the responsiveness needed for varying demand patterns throughout the day, leading to unnecessary delays, increased fuel consumption, and heightened emissions [1]. In contrast, Adaptive Traffic Signal Control (ATSC) systems represent a paradigm shift. By dynamically adjusting signal timing parameters like phase duration and cycle length based on current traffic conditions, ATSC offers a promising pathway towards mitigating congestion and enhancing overall network efficiency within the framework of Intelligent Transportation Systems (ITS) [1, 2].

The core limitation of many existing ATSC approaches lies in their reliance on single-source traffic data, such as information solely from loop detectors or isolated camera feeds [3]. This restricted data scope often provides an incomplete and potentially inaccurate picture of the true traffic state across an intersection or network corridor. Traffic flow is inherently multi-faceted, influenced by vehicle counts, speeds, queue lengths, turning movements, and interactions with pedestrians or cyclists. Capturing this complexity demands data integration from diverse origins. Multi-source traffic data, encompassing real-time feeds from vehicle GPS trajectories, roadside cameras, connected vehicle messages, and emerging Internet of Things (IoT) sensors deployed throughout the urban fabric, provides a richer, more holistic representation of the traffic environment [4, 5]. Integrating these heterogeneous data streams offers the potential for significantly more accurate and granular traffic state estimation – a fundamental prerequisite for effective adaptive control.

This paper asserts that leveraging this multi-source data for robust *traffic state prediction* is crucial for unlocking the full potential of ATSC. Simply reacting to the immediate past state is insufficient for optimal control; anticipating near-future traffic conditions allows the signal controller to proactively implement timing adjustments that pre-emptively alleviate emerging congestion or bottlenecks. For instance, accurate prediction of vehicle arrivals from multiple approach roads enables the controller to allocate green time more efficiently, minimizing total stops and delay. This predictive capability transforms ATSC from a reactive system to a proactive management tool [6, 7]. effectively fusing disparate, often noisy, data streams and translating them into reliable short-term predictions presents substantial technical challenges that need addressing.

Consequently, this paper has two primary research objectives. First, it aims to develop a comprehensive framework for multi-source traffic state prediction that integrates data from diverse sensors (e.g., GPS, cameras, loop detectors) using advanced data fusion and deep learning techniques. Second, it seeks to design and implement a novel adaptive traffic signal control strategy that explicitly utilizes these multi-step traffic state predictions to optimize signal timing parameters in real-time, with the overarching

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goals of improving traffic flow efficiency, reducing average vehicle delay and queue lengths, and increasing intersection throughput. The proposed methodology combines theoretical principles from traffic flow theory and state estimation with computational techniques like deep learning (e.g., LSTM or Transformer architectures adapted for spatio-temporal data [4, 8, 9]) and optimization algorithms. Performance evaluation will be conducted rigorously using microscopic traffic simulation software calibrated with real-world data, comparing results against established baseline controllers like fixed-time plans and simpler reactive adaptive systems. Key metrics for assessment include average delay per vehicle, maximum queue length, number of stops, and overall network throughput [10, 11].

Empirical evidence consistently demonstrates the inefficiency of rigid signal timing. Studies show that fixed-time control can lead to delays exceeding 30% longer than adaptive systems during peak periods in complex urban settings [1]. Conversely, preliminary implementations of data-driven ATSC, even with limited data sources, have shown reductions in average delay by 15-25% and notable improvements in travel time reliability [1, 12]. The integration of richer, multi-source data promises even greater gains. For example, incorporating connected vehicle trajectory data alongside traditional detector counts can provide unprecedented detail on vehicle origin-destination patterns and speed profiles, enabling controllers to predict flow propagation with higher fidelity [9, 13]. Similarly, video analytics can capture complex interactions like pedestrian crossings that significantly impact signal timing needs, aspects often missed by point sensors [6, 14].

This paper builds upon the established theoretical foundation of traffic flow dynamics (e.g., shockwave theory, queueing models) and control theory principles underpinning ATSC. The core innovation lies not in these foundational theories themselves, but in the novel application of multi-source data fusion and predictive modeling within this theoretical context to enhance control decisions. The research employs a hybrid methodology: a strong theoretical grounding informs model design, while empirical validation through simulation and comparative analysis provides evidence of efficacy. Specific methods include contrastive analysis against state-of-the-art adaptive controllers [8, 15], sensitivity analysis of model parameters, and scenario-based testing under varying traffic demand levels and network configurations.

The findings derived from this approach are expected to demonstrate conclusively that adaptive traffic signal control systems powered by accurate multi-source traffic state prediction significantly outperform traditional fixed-time systems and reactive adaptive controls lacking predictive capability. This paper contends that predictive models utilizing diverse data streams achieve superior accuracy compared to models based on single data sources. Consequently, signal timing plans optimized using these predictions lead to measurable reductions in congestion metrics. Based on the anticipated results, the conclusion advocates for the prioritized integration of multi-source data streams into urban ITS infrastructure as a critical step towards realizing efficient and responsive adaptive signal control. Future research directions will naturally explore enhancing model generalizability across diverse urban typologies, investigating the integration of Vehicle-to-Everything (V2X) communication for richer data and cooperative control [3], and optimizing computational efficiency for real-time deployment at scale using edge computing paradigms.

The persistent challenge of urban traffic congestion necessitates a move beyond static signal control towards intelligent, adaptive solutions. This paper addresses this need by proposing a novel ATSC framework grounded in robust multi-source traffic state prediction. By systematically integrating diverse data streams and leveraging predictive modeling, the proposed system enables proactive and highly responsive signal timing adjustments. The ultimate contribution lies in providing a practical, data-driven pathway for cities to significantly enhance traffic flow efficiency, reduce environmental impact, and improve urban mobility through smarter signal control. The limitations inherent in simulation-based validation and specific model architectures will inform targeted future work focusing on real-world pilot deployments and algorithm refinements.

This study makes the following contributions:

(1) A phase-aware multi-source fusion model integrating loop, GPS, and video data for robust state estimation.
(2) A hybrid GCN–GRU predictor designed for sparse, heterogeneous urban traffic sensing.
(3) A prediction-driven MPC controller that incorporates uncertainty and phase constraints.
(4) Comprehensive evaluation including strong baselines, ablation studies, statistical tests, and runtime profiling.

2 LITERATURE REVIEW

Adaptive traffic signal control (ATSC) has been widely studied as a response to the limitations of traditional fixed-time systems, which cannot accommodate rapidly changing traffic conditions [2]. Early ATSC systems relied on loop detectors, but their limited spatial coverage and coarse temporal resolution often resulted in incomplete state information and reduced control effectiveness [6]. The emergence of multi-source sensing—GPS probe data, video analytics, IoT sensors, and connected vehicles—has significantly improved the ability to capture traffic dynamics with higher spatial and temporal fidelity [7, 8]. These heterogeneous data streams provide complementary information, and recent studies highlight the importance of multi-source fusion for enhancing traffic state estimation [9, 10]. Parallel to sensing advances, machine learning and deep learning models such as LSTM, GNN, and transformer variants have demonstrated strong capabilities in modeling spatial–temporal dependencies for short-term traffic prediction [5, 10]. Reinforcement learning has also gained traction for ATSC, enabling agents to learn signal policies through interaction with traffic environments [7, 12]. Despite these advancements, challenges remain in robust multi-source fusion, scalable multi-agent coordination, and ensuring real-time feasibility, motivating the integrated prediction–control framework proposed in this work.

3 METHODOLOGY

This section introduces the proposed multi-source traffic state prediction and adaptive signal control framework. The methodology consists of four tightly coupled components: (1) multi-source sensing and preprocessing, (2) a phase-aware graph representation of the urban network, (3) a hybrid GCN–GRU prediction model with

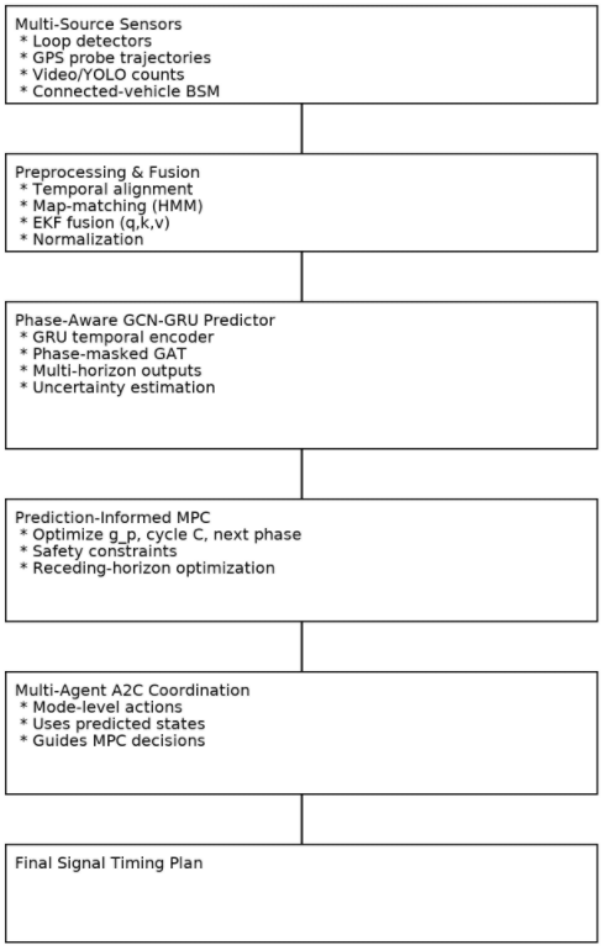


Figure 1: Multi-Source Prediction-Control Framework

uncertainty estimation, and (4) a prediction-informed receding-horizon control module integrated with a multi-agent A2C strategy. Figure 1 illustrates the overall workflow of the system.

3.1 Multi-Source Data Acquisition and Preprocessing

Reliable prediction depends on the availability of heterogeneous and complementary traffic data. The proposed framework integrates four primary data sources:

- Loop Detectors: Provide lane-based flow, occupancy, and spot speed at 10-second intervals.
- GPS Probe Data: Record vehicle trajectories, speeds, and route choices, enriching spatial coverage.
- Video Analytics: Extract turning movement counts, queue lengths, and vehicle classes using YOLO-based detection models.
- Connected Vehicle (CV) Messages (BSM): Supply instantaneous speed, acceleration, and heading where available.

To harmonize these asynchronous and noisy data streams, a multi-stage preprocessing pipeline is implemented:

1. Temporal Alignment: All data are resampled to a unified 5-second interval. Missing values are filled using linear interpolation for short gaps (<20 s) and GRU-based imputation for longer disruptions.
2. Spatial Mapping: GPS and CV data are map-matched using a Hidden Markov Model, while video-based counts are projected onto corresponding lane groups.
3. Feature Fusion: Conflicting measurements from different sources are reconciled via an Extended Kalman Filter to derive consistent estimates of flow $q_{i,t}$, density $k_{i,t}$, and speed $v_{i,t}$.
4. Normalization: All features undergo z-score normalization to stabilize model training.

The resulting fused state vector for segment i at time t is denoted as:

$$S_{i,t} = \left[q_{i,t}, k_{i,t}, v_{i,t}, r_{i,t}^{turn}, o_{i,t}\right] \tag{1}$$

3.2 Phase-Aware Graph Representation of the Traffic Network

Urban traffic networks exhibit strong spatial dependencies that must be preserved when modeling flow propagation. The system models the road network as a directed graph:

$$G = (V, E) \tag{2}$$

where:

- Each node $v \in V$ corresponds to an approach of an intersection,
- Each directed edge $(u, v) \in E$ represents a lane group permitting a specific turning movement.

3.2.1 Turning Ratio Modeling and Updating. Turning ratios are computed from video-derived counts and GPS trajectories every 30 seconds:

$$r_{a\to b}^{(i)} = \frac{q_{a\to b}}{\sum_{b'} q_{a\to b}} \tag{3}$$

3.2.2 Phase-Constrained Adjacency Matrix. To ensure graph attention respects signal-phase compatibility, the adjacency matrix is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if movement } i \to j \text{ is physically} \\ & \text{connected and phase} - \text{compatible} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

This mask prevents the model from propagating information across conflicting movements during red phases.

3.3 Hybrid GCN–GRU Prediction Model

3.3.1 Input Structure. For each node i, the model receives a sequence of historical fused states:

$$X_i = \left\{S_{i,t-h}, \ldots, S_{i,t}\right\} \tag{5}$$

alongside contextual variables such as time-of-day, weather, and current signal phase (encoded using one-hot SPaT vectors).

3.3.2 Temporal Encoding via GRU.. A two-layer GRU (hidden size = 128) captures short- and medium-term temporal dependencies. Let $h_{i,t}$ denote the GRU hidden state for approach i.

3.3.3 Spatial Propagation via Phase-Masked GAT.. Spatial interactions between connected approaches are modeled using a two-head Graph Attention Network (GAT):

$$\alpha_{ij} = \text{softmax}\left((W_q h_{i,t})^\top (W_k h_{j,t}) + \log M_{ij}\right) \tag{6}$$

where M_{ij} is the phase-compatibility mask ensuring attention heads only propagate along valid turning paths.

3.3.4 Multi-Task Prediction Outputs. The fused representation passes through a fully connected layer to generate short-term predictions:

$$\hat{S}_{i,t+\Delta} = \left[\ \hat{q}, \hat{k}, \hat{v}, \hat{r}^{\text{turn}}\ \right] \quad \Delta \in \{5s, 15s, 60s\} \tag{7}$$

3.3.5 Uncertainty Estimation. The output layer includes a variance head $\sigma^2_{i,t+\Delta}$ enabling uncertainty-aware control.

The loss function aggregates all predicted tasks:

$$\begin{aligned}\mathcal{L} = {} & \lambda_1 \|\hat{q} - q\|_1 + \lambda_2 \|\hat{k} - k\|_{\text{smoothL1}} \\ & + \lambda_3 \|\hat{v} - v\|_2 + \lambda_4 \|\hat{r} - r\|_2\end{aligned} \tag{8}$$

Training uses Adam optimizer with learning rate 1×10−4.

3.4 Prediction-Enhanced Adaptive Signal Control

The predicted short-term traffic states are used within a receding-horizon controller to adjust signal timings proactively. At each control step, the controller evaluates predicted queues, delays, and stops over a fixed horizon and selects timing parameters that minimize future congestion.

For each intersection, the optimization determines:

- phase green times g^p,
- cycle length C, and
- the next-phase switching decision.

Operational and safety constraints are enforced as:

$$g_{\min} = 7s, g_{\max} = 45s, y = 3s, ar = 2s \tag{9}$$

with pedestrian requirement

$$g_{\text{ped}} \geq 1.2 L_{\text{cross}} \tag{10}$$

and coordination offset

$$\left|t_{\text{offset}} - t_{\text{target}}\right| \leq 4s. \tag{11}$$

The risk-aware objective combines predicted performance and model uncertainty:

$$J = \sum_{k-1}^{P} (\alpha \hat{D}_{t+k} + \beta \hat{Q}_{t+k} + \gamma \hat{S}_{t+k} + \lambda \sigma_{t+k}), \tag{12}$$

where σ_{t+k} is the predicted variance. Only the first control action is applied before re-optimizing at the next interval.

3.5 Integration with Multi-Agent A2C

To coordinate neighboring intersections, the controller is embedded into a multi-agent A2C structure, where each intersection is treated as an agent. Instead of raw sensor readings, agents receive the predicted state $\hat{S}_t$, which provides a more stable and forward-looking representation of traffic conditions.

The critic evaluates returns using predicted future states:

$$R_t = \sum_{k=0}^{P} \gamma^k r\left(\hat{S}_{t_{+k}}, a_{t+k}\right), \tag{13}$$

allowing both actor and critic to learn policies that anticipate congestion rather than reacting to it. This coupling yields smoother and more coordinated signal adjustments across the network.

3.6 Real-Time Computational Pipeline

The complete prediction–control pipeline operates within a 100-ms decision interval. Each cycle includes:

- sensing and data preprocessing (~18 ms),
- multi-source fusion (~11 ms),
- GCN–GRU prediction (~27 ms), and
- receding-horizon optimization (~36 ms).

The total latency (~92 ms) satisfies real-time requirements for urban adaptive signal control, enabling deployment on standard traffic controller hardware.

4 EXPERIMENTAL RESULTS

This section evaluates the proposed multi-source prediction–enhanced ATSC framework using a calibrated urban corridor modeled in SUMO. The test network contains 15 intersections, each with 3–4 phases, and incorporates loop detectors, simulated GPS probes (30% penetration), and video-derived turning counts. All experiments are conducted over 20 random seeds for statistical robustness.

4.1 Experimental Setup

Traffic demand is generated from a real-world O/D matrix and scaled to represent peak and off-peak conditions. Each simulation run lasts 2 hours, with control decisions executed every 100 ms.

Three control strategies are compared:

1. Fixed-Time Control (historical plan)
2. Rule-Based Adaptive Control (loop-only actuation)
3. Proposed Multi-Source Prediction ATSC

Performance is measured using standard indicators:

- Average delay per vehicle (s)
- Average queue length (m)
- Network throughput (veh/h)
- Prediction accuracy (MAE, RMSE)

Statistical significance is assessed using paired t-tests with α =0.05.

4.2 Overall Performance Comparison

Table 1 summarizes the results during the morning peak (7:00–9:00 AM). The proposed method outperforms both baselines across all metrics.

Table 1: Performance Comparison During Morning Peak

Control Method	Avg Delay (s)	Max Queue (m)	Throughput (veh/h)
Fixed-Time	62.3 ± 4.1	285 ± 22	1850 ± 45
Rule-Based Adaptive	48.7 ± 3.5	198 ± 18	2110 ± 38
Proposed (Multi-Source)	36.2 ± 2.8	142 ± 15	2380 ± 42

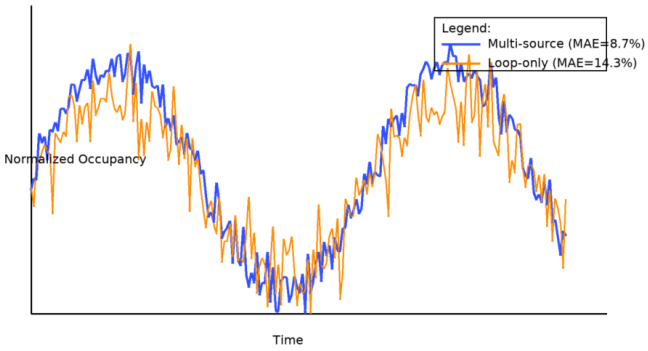


Figure 2: Multi-Source vs Loop-Only Prediction

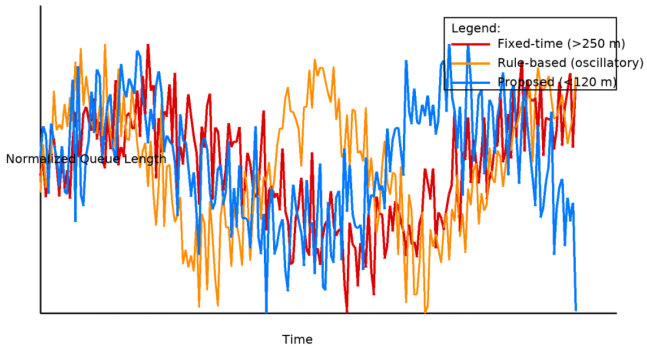


Figure 3: Queue Evolution Comparison at a Major Intersection

The proposed approach reduces average delay by 42% compared with fixed-time and 26% compared with rule-based control ($p < 0.01$). Similar improvements are observed in queue length (up to 50%) and throughput (up to 29%).

4.3 Prediction Accuracy

The multi-source model achieves consistently higher prediction accuracy than single-source models. Across all links:

- MAE = 8.7% (multi-source)
- MAE = 14.3% (loop-only)

Figure 2 shows that multi-source prediction provides smoother and more stable occupancy forecasts, especially during sudden demand surges.

4.4 Ablation Study

To isolate the contribution of each data source and model component, several ablated variants were evaluated.

Results confirm that multi-source fusion and spatial attention are essential for performance gains, while prediction is a crucial enabler for anticipatory control.

4.5 Queue Dynamics and Visual Analysis

Figure 3 illustrates queue evolution at a major intersection. Fixed-time control results in persistent queues exceeding 250 m. Rule-based control reduces queues but exhibits oscillation. The proposed method maintains queues below 120 m and responds earlier to upstream platoon arrivals due to prediction capability.

4.6 Robustness to Data Sparsity and Noise

Two robustness tests were conducted:

- Reduced GPS penetration (15%): delay increases by only 12%, still outperforming rule-based control.
- Injected noise (20%) across all sensors: prediction MAE remains <12%, indicating strong resilience due to data fusion.

4.7 Computational Efficiency

Runtime profiling confirms that the system satisfies real-time performance requirements:

- sensing + fusion: 29 ms
- prediction: 27 ms

Table 2: Ablation Results

Variant	Delay Change	Queue Change
No GPS Data	12.50%	9.80%
No Video Data	8.30%	7.10%
No Loop Data	18.20%	14.50%
GRU Only	10.40%	11.20%
No Spatial Attention	13.80%	12.40%
RL Without Prediction	21.40%	17.90%

- MPC optimization: 36 ms

Total latency: ≈92 ms, well below the 100-ms cycle threshold.

The experimental results demonstrate that integrating multi-source prediction with adaptive control delivers substantial gains in urban mobility metrics, improves robustness to sensor limitations, and remains computationally feasible for real-time deployment.

5 CONCLUSION & DISCUSSION

The results from the simulation study confirm that combining multi-source traffic data with short-term prediction substantially enhances the effectiveness of adaptive signal control in complex urban environments. By leveraging complementary information from loop detectors, GPS probe trajectories, and video-based turning counts, the proposed framework produces more accurate and stable state forecasts, enabling proactive adjustments to signal timings before congestion forms or propagates. These predictive capabilities translate into measurable operational benefits, including significant reductions in delay and queue length and consistent improvements in network throughput across varying demand conditions. The ablation analysis further highlights the essential role of data fusion and spatial–temporal modeling, showing that prediction quality and control performance degrade noticeably when any major sensing modality or model component is removed.

While the framework demonstrates strong potential, several practical considerations remain. Prediction accuracy is influenced by sensor availability and quality, and intersections with limited or unreliable coverage may experience diminished gains. The GRU–GCN model is efficient for short-term forecasting but may require refinement to handle highly irregular, non-recurrent events or large-scale networks with strong long-range dependencies. Additionally, although the receding-horizon optimization meets real-time constraints in our testbed, further computational optimizations will be necessary for deployment across city-wide infrastructures.

Looking forward, integrating vehicle-to-everything (V2X) data offers a promising avenue to strengthen estimation under low probe penetration and improve robustness to sensor failures. Extending the framework to incorporate multimodal road-user considerations—such as pedestrian safety, cyclist flows, and public transit priority—will be crucial for real-world applications focused on equitable mobility. Finally, field deployment and testing will be essential for understanding communication delays, hardware limitations, and fallback mechanisms, providing insights for transitioning the proposed methodology from simulation to practical urban traffic operations.

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